fn sample_logarithmic_exp

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This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `fn sample_logarithmic_exp` in `mod.rs`. This proof uses the exact geometric-exp sampler and exact rational Bernoulli sampler to realize the logarithmic distribution.

1 Hoare Triple

Precondition

$\mathtt{x} \in \mathbb{Q} \wedge \mathtt{x} > 0$

Pseudocode

```
def sample_logarithmic_exp(x):
    while True:
        k = sample_geometric_exp_fast(x)
        if k == 0:
            continue

        if sample_bernoulli_rational(1 / k):
            return k
```

Postcondition

Theorem 1.1. For any setting of the input parameter `x` such that the given preconditions hold, `sample_logarithmic_exp` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed on $\mathbb{N}_{>0}$ with

$$\forall k \in \mathbb{N}_{>0}, \qquad \Pr[\mathtt{out} = k] = \frac{e^{-\mathtt{x}k}}{k \log(1/(1 - e^{-\mathtt{x}}))}.$$

2 Proof

Assume the preconditions are met and write $\lambda = e^{-x} \in (0, 1)$.

Lemma 2.1. `sample_logarithmic_exp` only returns `Err(e)` when there is a lack of system entropy.

Proof. Since $\mathtt{x} > 0$, the precondition on `sample_geometric_exp_fast` is met. By its definition, it can only return an error due to a lack of system entropy.

If `k` on line **??** is nonzero, then `k` is a positive integer, so the preconditions on `sample_bernoulli_rational` are met when called with probability $1/\mathtt{k}$ on line **??**. By its definition, it can only return an error due to a lack of system entropy.

The only source of errors in `sample_logarithmic_exp` is from the invocation of these functions, therefore `sample_logarithmic_exp` only returns `Err(e)` when there is a lack of system entropy. □

We now condition on not returning an error and fix any iteration of the loop.

- Let g be a realization of a random variable $G \sim \mathit{Geometric}(1 - e^{-x})$, supported on $\mathbb{N}_0$.
- If $g > 0$, let b be a realization of a random variable $B \sim \mathit{Bernoulli}(1/g)$.

Lemma 2.2. For any $k \in \mathbb{N}_{>0}$, the probability that a single iteration of the loop returns k is

$$(1 - e^{-x})\frac{e^{-xk}}{k}.$$

Proof. A single iteration returns k exactly when:

1. the geometric draw equals k, and
2. the Bernoulli draw with success probability $1/k$ succeeds.

Therefore,

$$\begin{aligned}\Pr[\text{single iteration returns } k] &= \Pr[G = k]\Pr[B = \top \mid G = k] \\ &= (1 - e^{-x})e^{-xk} \cdot \frac{1}{k}.\end{aligned}$$

□

Lemma 2.3. The probability that a single iteration of the loop returns some value in $\mathbb{N}_{>0}$ is

$$(1 - e^{-x}) \log\left(\frac{1}{1 - e^{-x}}\right).$$

Proof. By Lemma 2.2,

$$\begin{aligned}\Pr[\text{single iteration returns a value}] &= \sum_{k=1}^{\infty}(1 - e^{-x})\frac{e^{-xk}}{k} \\ &= (1 - e^{-x})\sum_{k=1}^{\infty}\frac{(e^{-x})^k}{k}.\end{aligned}$$

Since $e^{-x} \in (0, 1)$, the power-series identity

$$\sum_{k=1}^{\infty}\frac{z^k}{k} = \log\left(\frac{1}{1 - z}\right) \quad \text{for } z \in (0, 1)$$

applies with $z = e^{-x}$, giving

$$\Pr[\text{single iteration returns a value}] = (1 - e^{-x}) \log\left(\frac{1}{1 - e^{-x}}\right).$$

□

Lemma 2.4. If the outcome of `sample_logarithmic_exp` is `Ok(out)`, then `out` is distributed on $\mathbb{N}_{>0}$ with

$$\forall k \in \mathbb{N}_{>0}, \qquad \Pr[\texttt{out} = k] = \frac{e^{-xk}}{k \log(1/(1 - e^{-x}))}.$$

Proof. By Lemma 2.2, the probability that a fixed loop iteration returns k is

$$r_k := (1 - e^{-x}) \frac{e^{-xk}}{k}.$$

By Lemma 2.3, the probability that a fixed loop iteration returns any value is

$$a := (1 - e^{-x}) \log\left(\frac{1}{1 - e^{-x}}\right).$$

Since loop iterations are independent, the probability that the algorithm returns k on the t-th iteration is

$$(1 - a)^{t-1} r_k.$$

Summing over all $t \geq 1$,

$$\begin{aligned}
\Pr[\texttt{out} = k] &= \sum_{t=1}^{\infty} (1 - a)^{t-1} r_k \\
&= \frac{r_k}{a} \\
&= \frac{(1 - e^{-x}) e^{-xk}/k}{(1 - e^{-x}) \log(1/(1 - e^{-x}))} \\
&= \frac{e^{-xk}}{k \log(1/(1 - e^{-x}))}.
\end{aligned}$$

Thus `out` has the claimed distribution. □

Proof of Theorem 1.1. Holds by Lemma 2.1 and Lemma 2.4. □